

Day 13 Notes – AI Models & LLMs

AI Models

An **AI Model** is a computer program that learns patterns from data and uses those patterns to make predictions, answer questions, generate content, or solve problems.

Examples

- ChatGPT
- Google Gemini
- Claude
- Llama
- Mistral

AI models are trained using huge amounts of data so they can understand different tasks.

What are LLMs?

LLM = Large Language Model

A **Large Language Model (LLM)** is an AI model trained on **massive amounts of text data** (books, websites, articles, research papers, conversations, etc.) to understand and generate human language.

Definition

A Large Language Model (LLM) is an AI model trained on large-scale text data to understand, process, and generate human-like text.

What Can an LLM Do?

LLMs can perform many Natural Language Processing (NLP) tasks such as:

- Answer questions
- Generate code
- Summarize documents

- Translate languages
- Write emails
- Generate stories
- Explain concepts
- Solve programming problems
- Chat with users

Popular LLMs

Company Model

OpenAI GPT-4, GPT-5

Google Gemini

Meta Llama

Anthropic Claude

Mistral AI Mistral

DeepSeek DeepSeek

xAI Grok

How does an LLM work?

Step 1: Training

The model is trained using billions or trillions of words from:

- Books
- Wikipedia
- Research papers
- Websites
- Public code repositories

During training it learns:

- Grammar
- Vocabulary
- Facts
- Programming languages
- Writing styles
- Relationships between words

Step 2: Prompt

The user gives a prompt.

Example:

Explain Java OOP concepts.

Step 3: Processing

The LLM understands the context using neural networks (Transformer architecture).

Step 4: Output

The model generates the most likely next words to produce a useful response.

Why are LLMs called "Large"?

Because they have:

- Huge datasets
- Billions of parameters
- Massive computing power
- Large memory requirements

Example:

GPT models contain billions or even trillions of parameters.

Applications of LLMs

- Chatbots
- Virtual Assistants
- Code Generation
- Customer Support
- Content Writing
- Education
- Healthcare
- Finance
- Research
- Translation

AI Platforms

AI platforms provide access to AI models through APIs.

Popular platforms include:

1. **Google AI Studio**
2. **OpenAI Platform**
3. **Groq Platform**

1. Google AI Studio

Google AI Studio allows developers to use **Gemini models**.

Features

- Free API key (with limits)
- Prompt testing

- Chat interface
- Image understanding
- Code generation
- Supports Gemini models

Uses:

- Chatbots
- AI apps
- Summarization
- Code assistants

2. OpenAI Platform

The OpenAI Platform provides APIs for GPT models.

Features

- GPT models
- Image generation
- Speech-to-text
- Text-to-speech
- Embeddings
- Function calling

Uses:

- AI assistants
- Chat applications
- Coding assistants
- Business automation

3. Groq Platform

Groq provides **very fast inference** for open-source AI models.

Important: Groq does **not** create its own LLMs. It hosts and serves models such as:

- Llama
- Gemma
- DeepSeek
- Mistral
- Qwen

The main advantage is **high-speed response times**.

Rate Limit

A **Rate Limit** is the maximum number of API requests you can make within a certain time period.

Example:

Allowed:

- 30 requests/minute

If you exceed this limit:

 Error:

429 Too Many Requests

Purpose:

- Prevent server overload
- Ensure fair usage
- Protect APIs from abuse

Token Limit

AI models don't read words directly. They process **tokens**.

A **token** is a small unit of text.

Example:

Sentence:

I love Java programming.

Tokens might be:

"I"

"love"

"Java"

"programming"

."

Approximately:

- 1 token $\approx$ 0.75 English words
- 100 words $\approx$ 130 tokens (rough estimate)

Context Window

The token limit includes:

- Your prompt
- Previous conversation
- AI's response

Example:

Model Context Window = 8,000 tokens

Prompt:

3000 tokens

Conversation history:

2000 tokens

Maximum response:

3000 tokens

Total:

$3000 + 2000 + 3000 = 8000$ tokens

Difference between Rate Limit and Token Limit

Rate Limit	Token Limit
Limits number of requests	Limits amount of text processed
Measured per minute/day	Measured in tokens
Prevents excessive API calls	Prevents overly large inputs/outputs